

Supplementary Data S14: List of 11 species with specimen identifications found in the Madrean Sky Islands that need further examination. These observations fall outside of the currently understood ranges of these species, implying either the range should be expanded or these specimens have been misidentified. Both *P. truei* and *P. difficilis* are known to occur on specific sky islands, however, some specimens have been reported outside of these known mountains, warranting additional verification. Specimens of *L. ega* and *M. keenii* likely belong to *L. xanthinus* and *M. evotis* respectively, but verification of specimens is also needed.

Order	Species	Reason	Evidence
Rodentia	<i>Ammospermophilus leucurus</i>	Out of range	Belk, M. C., & Smith, H. D. (1991). <i>Ammospermophilus leucurus</i> . Mammalian Species, 368, 1–8. https://doi.org/10.2307/3504191
Rodentia	<i>Peromyscus beatae</i>	Out of range	Álvarez-Castañeda, S. T., Álvarez, T., & González-Ruiz, N. (2017). Keys for Identifying Mexican Mammals. JHU Press. Tiemann-Boege, I., Kilpatrick, C. W., Schmidly, D. J., & Bradley, R. D. (2000). Molecular Phylogenetics of the <i>Peromyscus boylii</i> Species Group (Rodentia: Muridae) Based on Mitochondrial Cytochrome b Sequences. Molecular Phylogenetics and Evolution, 16(3),

			366–378. https://doi.org/10.1006/mpev.2000.0806
Rodentia	<i>Peromyscus californicus</i>	Out of range	Merritt, J. F. (1978). <i>Peromyscus californicus</i> . <i>Mammalian Species</i> , 85, 1–6. https://doi.org/10.2307/3503909
Rodentia	<i>Peromyscus difficilis</i>	Out of range	Fernández, J. A., García-Campusano, F., & Hafner, M. S. (2010). <i>Peromyscus difficilis</i> (Rodentia: Cricetidae). <i>Mammalian Species</i> , 42(1), 220–229.
Rodentia	<i>Peromyscus truei</i>	Out of range	Hoffmeister, D. F. (1981). <i>Peromyscus truei</i> . <i>Mammalian Species</i> , 161, 1–5. https://doi.org/10.2307/3503851 Cook, J. (1986). The Mammals of the Animas Mountains and Adjacent Areas, Hidalgo County, New Mexico. <i>Occasional Papers</i> . https://digitalrepository.unm.edu/occasionalpapers/11
Rodentia	<i>Thomomys talpoides</i>	Out of range	Verts, B. J., & Carraway, L. N. (1999). <i>Thomomys talpoides</i> . <i>Mammalian Species</i> , 618, 1–11.

			https://doi.org/10.2307/3504451
Chiroptera	<i>Lasiurus ega</i>	Out of range, taxonomic dispute	Kurta, A., & Lehr, G. C. (1995). <i>Lasiurus ega</i>. Mammalian Species. Mammal Diversity Database. (2025). Mammal Diversity Database (Version 2.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.15007505
Chiroptera	<i>Leptonycteris curasoae</i>	Out of range	Cole, F. R., & Wilson, D. E. (2006). <i>Leptonycteris curasoae</i>. Mammalian Species, 796, 1–3. https://doi.org/10.1644/796.1
Chiroptera	<i>Myotis keenii</i>	Out of range, taxonomic dispute	Fitch, J. H., & Shump, Jr., K. A. (1979). <i>Myotis keenii</i>. Mammalian Species. Morales, A. E., Fenton, M. B., Carstens, B. C., & Simmons, N. B. (2021). Comment on “Population genetics reveal <i>Myotis keenii</i> (Keen’s myotis) and <i>Myotis evotis</i> (long-eared myotis) to be a single species.” Canadian Journal of Zoology, 99(5), 415–422. https://doi.org/10.1139/cjz-2020-0048

Chiroptera	<i>Myotis leibii</i>	Out of range	Best, T. L., & Jennings, J. B. (1997). <i>Myotis leibii</i> . Mammalian Species, 547, 1–6. https://doi.org/10.2307/3504255
Eulipotyphla	<i>Sorex vagrans</i>	Out of range	Gillihan, S. W., & Foresman, K. R. (2004). <i>Sorex vagrans</i> . Mammalian Species, 744, 1–5.